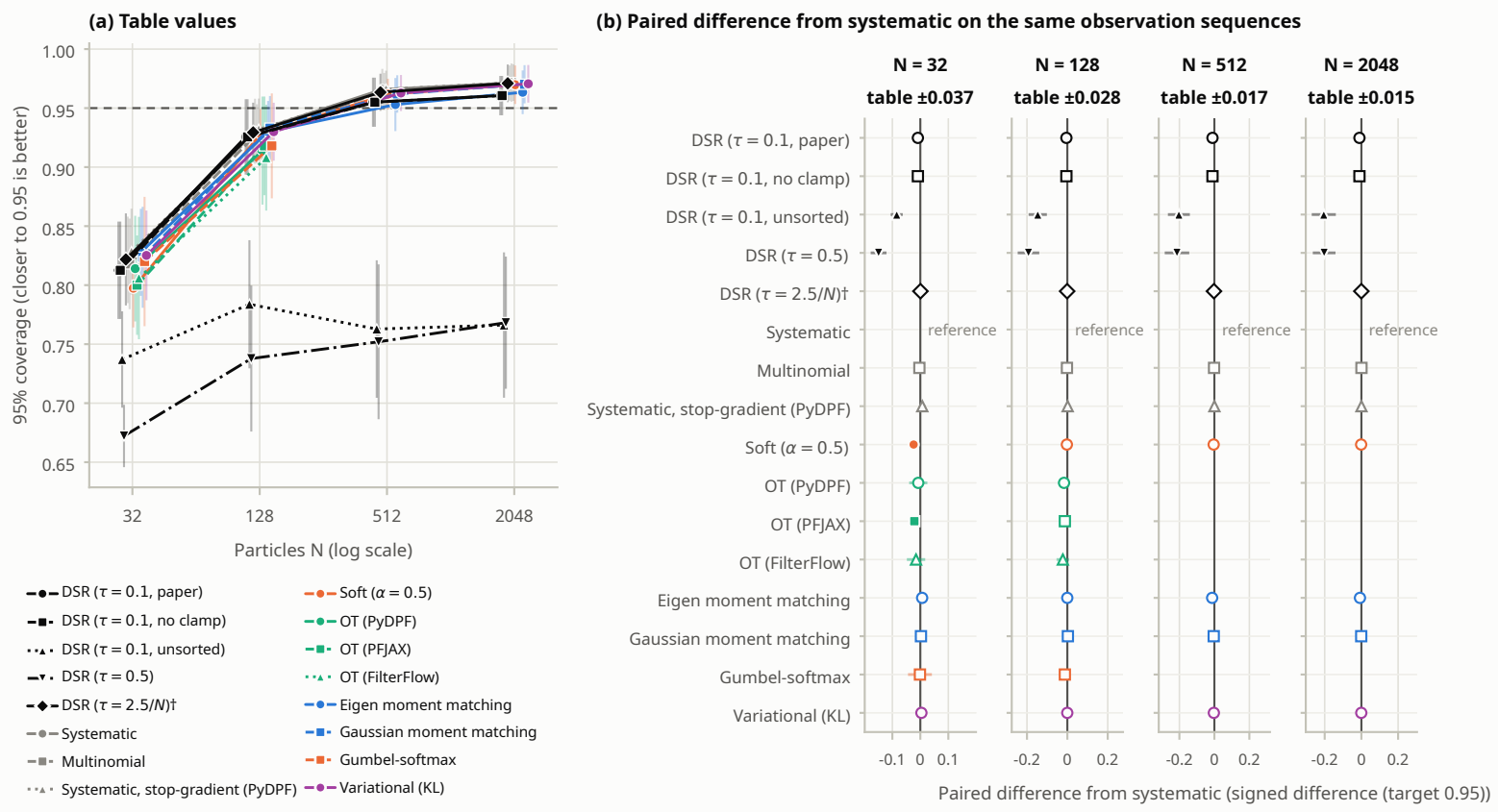

(a) Table values
95% coverage (closer to 0.95 is better)
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
32
128
512
2048
Particles N (log scale)
DSR (τ = 0.1, paper)
DSR (τ = 0.1, no clamp)
DSR (τ = 0.1, unsorted)
DSR (τ = 0.5)
DSR (τ = 2.5/N)†
Systematic
Multinomial
Systematic, stop-gradient (PyDPF)
Soft (α = 0.5)
OT (PyDPF)
OT (PFJAX)
OT (FilterFlow)
Eigen moment matching
Gaussian moment matching
Gumbel-softmax
Variational (KL)
(b) Paired difference from systematic on the same observation sequences
N = 32
table ±0.037
N = 128
table ±0.028
N = 512
table ±0.017
N = 2048
table ±0.015
reference
-0.1 0 0.1
-0.2 0 0.2
Paired difference from systematic (signed difference (target 0.95))